

■ exposure ■ treatment proxy Z

■ outcome proxy W

Molecular panel

$X: n$ patients
 $\times p$ variables

Niche index

$s_k = \sum_j X_{kj}$
patients sorted

Power curves

$c_j(s) = a_j s^{b_j}$

Deviations

$U = X - C(s)$

Latent state

$U \approx F\Lambda'; r, \rho_i$

Weak-form ODE

$\dot{x}_j = Q_{jj}(x_j) + \sum_k Q_{jk}(x_k)$

Dynamic network

signed, weighted,
directed

Exposure, outcome

A : a column of X
 Y : clinical outcome

Λ, r

quasi-dynamic context

Support graph

nodewise LASSO
 k niche windows

Proxy construction

Z : adjacent to A
 W : outside A component

Proximal bridge

$E[Y|A, Z]$
 $= E[h(W, A) | A, Z]$

Effect estimation

$\hat{\beta}; \nu$ (proxy strength)
linear bridge / Cox extension